The remaining mixed orbit defect

Human Compression project: bounded L044 proof component

Retain the accepted unique-crossing setting in a transitive string C-group $G \leq A_n$, $n \geq 12$, with every maximal omission intransitive. Its two G_i-orbits A, B have maximal block systems $(k_A, m_A) = (k, 2)$ and $(k_B, m_B) = (2, m)$, $m \geq 3$, and crossing endpoints a, b. Assume $\alpha = 0$, $\beta = 1$, $\gamma = 0$. Here M is the minimal distinguished subfamily generating the simultaneous block image, with the accepted actual-orbit deficits. Then $|M| = m$ and $r = k + m$. Write $L = \langle \rho_j : j < i \rangle$ and $R = \langle \rho_j : j > i \rangle$, so $G_i = LR$ and L, R commute. We distinguish block images from their actual permutation lifts throughout.

Lemma 1 (The block-image marking)**.** *After reversal, M consists of $\tau = \rho_{i-1}$ and the interval $h_0, \ldots, h_{m-2} = \rho_{i+1}, \ldots, \rho_{i+m-1}$. The image of τ is pure C_2 on the two blocks of A; the other images form a pure maximal anchored S_m marking on the blocks of B. These are image statements; the actual group $H = \langle h_0, \ldots, h_{m-2} \rangle$ may have a block kernel.*

Proof. The independent block-image family has length m, so its group is the full $C_2 \times S_m$ by the paid normal-quotient and Whiston equality argument. Repeat the block-image part of Lemma 1 of the accepted mixed-block note: a minimal subfamily generating S_m has $m - 1$ members forming a connected extremal string interval next to i. The one remaining generator has trivial S_m image, by the primitive interval lemma on the opposite side, or the anchored endpoint-omission centralizer on the same side. It is the pure C_2 element and must have index $i - 1$.

This argument does not assume IP in the image. In the next step of the earlier proof, replace the use of IP by minimality of M in its simultaneous block image: if the subgroup generated by the S_m interval contained the pure C_2 element, that remaining generator would be redundant in the block image. Thus the interval is a character graph over S_m. Its tail fixes the crossing block and generates the natural S_{m-1}; some tail image is odd. Since all tail indices are distant from i, they have trivial binary image. The character is therefore trivial. This proves the statement entirely in the block image, without asserting that its lift is faithful. □

The groups $\langle \tau \rangle$ and H commute by their actual indices. Every M-orbit in A has even size, and every M-orbit in B meets every one of its m blocks. The single orbit deficit therefore has precisely two possible locations:

1. M has one four-point orbit D and $k-2$ paired orbits in A, and two natural m-point copies in B;

2. M has k paired orbits in A and is transitive on all $2m$ points of B.

Lemma 2 (The defect on the two-block orbit)**.** *The first alternative is impossible.*

Proof. On B, the two M-orbits meet each block in one point. The action there is faithfully the diagonal natural S_m, and τ is trivial there. On every paired M-orbit in A, H is trivial because its binary block image is trivial. On the unique orbit D, τ exchanges the two blocks, while H preserves them and commutes with τ. Thus the image of H on D is the diagonal C_2 on its two within-block pairs. Denote its nonidentity element by z, extended by the identity outside D.

If $a \notin D$, every right generator fixes a, so the commuting anchor lemma makes the whole right interval trivial on A. This contradicts the nontrivial H-action on D. Hence $a \in D$. Only

h_0 can act nontrivially on D: each later h_j fixes a, and a nonidentity element of this diagonal C_2 has no fixed point on D.

Consequently the actual H embeds in $\langle z\rangle \times S_m$. Its tail has trivial z coordinate and generates the natural S_{m-1} on the nonanchor blocks of B. The first member has nontrivial z coordinate. If $z \notin H$, this subgroup would be a character graph over S_m, impossible because the tail contains an odd image but has trivial z coordinate. Thus the pure kernel element z really belongs to H. This is a proved kernel occurrence, not a conclusion drawn from block-image notation.

The entire left interval commutes with H and hence preserves the support D of z. The right generators in H preserve D by construction. Every right generator beyond H fixes a and commutes with h_0 and τ: the first possible such index is $i+m$, at distance $m-1 \geq 2$ from h_0. The group $\langle \tau, h_0\rangle$ is transitive on D, so those further generators fix D pointwise. Thus G_i preserves D, contradicting transitivity on A unless $k=2$. When $k=2$, both original systems have two-point blocks; this is already excluded in the present rank-failure range by the complete accepted two-point-block lemma. Its range applies because $n \geq 12$ then forces $m \geq 4$. □

Lemma 3 (The defect on the two-point-block orbit)**.** *The second alternative is impossible.*

Proof. The group H fixes A pointwise: all paired M-orbits project bijectively to the two blocks, and H has trivial binary image. Every further right generator fixes a too. The commuting anchor lemma therefore makes the entire right interval trivial on A. The left interval is transitive on A, though its action on B need not be trivial.

The block image of H on B is transitive S_m, so its point action is either transitive on all $2m$ points, or has two m-point orbits, each a faithful natural S_m copy. In either case an element centralizing H and fixing b is the identity on B: use transitivity in the first case; in the second, fixing one point fixes that whole copy, and the other natural copy has trivial internal centralizer. Thus every left distinguished generator except τ is trivial on B, because it centralizes H and fixes b. We do not impose this on τ, which may perform the uniform within-block flip on B.

There is no right generator beyond the H interval. Any generator more distant than its one outer neighbor centralizes H and fixes b, hence is trivial on B and already trivial on A. The one possible outer neighbor $v = \rho_{i+m}$ centralizes the far-end omission of H. Its block image fixes the crossing block and is trivial by the anchored symmetric centralizer lemma. It fixes the crossing two-point block pointwise, and commutation propagates this fixation to at least $m-1$ blocks. It can therefore move at most the remaining two-point block, by a single transposition. Since it is trivial on A, evenness excludes that as well. The entire right interval is exactly H, of rank $m-1$.

The left interval consequently has rank k, since $r = k+m$. It ends in $x = \rho_{i-2}$ and τ, with x present to connect the paired anchor to another row. Its preceding $k-2$ generators are trivial on B, commute with τ, and fix a. They act faithfully and diagonally on the $k-1$ nonanchor rows of A, so their independent marking generates the full S_{k-1} by Whiston. The generator x is also trivial on B.

For $k \geq 5$, the accepted maximal symmetric-string endpoint classification excludes this equality. For completeness, in an ordinary marking the near-end omission is S_{k-2}, whose centralizer on a block is the transposition of the anchor and one unused row. Fixation of a makes x trivial on its block and a single odd transposition on the other, impossible. The degree-four twisted marking ($k=5$) and outer degree-six marking ($k=7$) can instead have a transitive endpoint omission on all nonanchor rows. Any permutation centralizing it preserves its unique fixed point, so x fixes the opposite anchor, again impossible. These are exactly the previously accepted small/outer endpoint cases; no new small-group classification is used.

For $k=2$, the same argument is immediate on two-point blocks: x is one odd transposition. For $k=3$, label the rows $0, 1, 2$ with $a=0$ and their mates $0', 1', 2'$. The sole preceding generator is $p = (1\ 2)(1'\ 2')$. Since x fixes 0, moves $0'$ to another row and is even, it is

$$x = (1\ 2)(0'\ j'), \qquad j \in \{1, 2\}.$$

The group $\langle x, \tau \rangle$ is transitive on the six points: contract the three τ pairs, and its row edges are 1–2 and 0–j, a spanning tree. At $k = 4$ the complete accepted endpoint calculation gives

$$p = (1\ 2)(1'\ 2'), \quad q = (2\ 3)(2'\ 3'), \quad x = (1\ 2)(0'\ 3') \text{ or } (1'\ 2')(0'\ 3').$$

Then $\langle q, x, \tau \rangle$ is transitive on the eight points, by the accepted row path 0–3–2–1.

In either small case, omit the far generator p from G. Its remaining generators are transitive on A. They still contain $\langle \tau, H \rangle = \langle M \rangle$, which by this case's hypothesis is transitive on B, whether or not H alone is transitive there. The generator ρ_i joins the two orbits. Thus this global maximal omission is transitive, the final contradiction. □

Boundary and attribution. This completes the last numerical mixed case in the accepted exact-deficit reduction. Together with the complete two-point-block, two-block and mixed beta-zero components, it proves the rank bound $r \leq \lfloor (n-1)/2 \rfloor$ of L044 Proposition 5.7 in the standing setting. The source framework is Cameron–Fernandes–Leemans–Mixer, arXiv:1605.09173v1, Section 5.1; the complete paid generation, anchor and endpoint proofs remain explicit dependencies. The finer source block diagrams, other main source branches and exact tables still require their own dispositions. No full-original recovery, new census, Lean theorem or question credit is claimed.